# CROP OUTLOOK · KALRO Kiboko, Makueni Farm

Kenya, Short Rains 2026 · experimental · AquaCrop (the published outlook) and WOFOST (a second crop model) grown in every member of ECMWF SEAS5 (Oct 01 + latest obs), observations to 5 Nov; crops sown by FAO agricultural onset

## The season so far

Rain since 1 Oct: 44 mm (98% of normal), CHIRPS to 5 Nov; soil water 58% of the available water on 31 Oct (FAO water balance).

## Outlook by crop

| Crop | Sowing (median) | Yield if sown AquaCrop (8 in 10) | Yield if sown WOFOST | Failure if sown AquaCrop / WOFOST | Past seasons yield / failure | Crop models |
|---|---|---|---|---|---|---|
| Early-maturing maize | 29 Oct | 2.9 t/ha (0–5.6) | 2.2 t/ha | 24% / 12% | 0.8 t/ha / 48% | agree |
| Medium-maturing maize | 29 Oct | 3.1 t/ha (0–6.7) | 3.2 t/ha | 24% / 12% | 0.4 t/ha / 64% | agree |
| Sorghum | 29 Oct | 3.3 t/ha (0–4.7) | 3.1 t/ha | 24% / 8% | 0.9 t/ha / 48% | partly agree |
| Dry bean | 29 Oct | 1.3 t/ha (0–2.0) | no WOFOST | 16% / — | 0.7 t/ha / 27% | — |

Yield if sown: median of the sown crops (8 in 10 in brackets); failure: a yield below the crop's threshold (0.5 t/ha, bean 0.2). Simulated past seasons: the same model in the observed weather of 1993–2025 (the normal). Crop models: agree when the yields are within 1 t/ha and the failure chances within 15 points; depends on the model beyond twice that. Water-limited yields (no nutrient, pest or management losses): an upper bound for most fields.

## What management could change (change in median yield, AquaCrop / WOFOST)

### Early-maturing maize: the largest gain short of full irrigation is supplementary irrigation, up to 100 mm a season (+2.4 t/ha AquaCrop, +2.3 WOFOST)

| Choice | AquaCrop | WOFOST |
|---|---|---|
| Sow 14 days earlier | -1.0 t/ha | -1.5 t/ha |
| Sow 14 days later | -1.0 t/ha | -0.3 t/ha |
| Supplementary irrigation, up to 100 mm a season (100 mm) | +2.4 t/ha | +2.3 t/ha |
| Mulch, 50 % residue cover | +1.2 t/ha | +0.8 t/ha |
| Tied ridges or pits (no surface runoff) | +0.7 t/ha | +0.3 t/ha |
| Plant density × 1.2 | +0.2 t/ha | — |
| Full irrigation (no water stress) (206 mm) | +2.9 t/ha | +3.6 t/ha |

### Sorghum: the largest gain short of full irrigation is supplementary irrigation, up to 100 mm a season (+1.3 t/ha AquaCrop, +0.9 WOFOST)

| Choice | AquaCrop | WOFOST |
|---|---|---|
| Sow 14 days earlier | -1.8 t/ha | -0.3 t/ha |
| Sow 14 days later | -1.9 t/ha | -0.6 t/ha |
| Supplementary irrigation, up to 100 mm a season (75 mm) | +1.3 t/ha | +0.9 t/ha |
| Mulch, 50 % residue cover | +0.3 t/ha | -0.0 t/ha |
| Tied ridges or pits (no surface runoff) | +0.2 t/ha | +0.0 t/ha |
| Plant density × 1.2 | +0.1 t/ha | — |
| Full irrigation (no water stress) (176 mm) | +1.5 t/ha | +1.6 t/ha |

Each choice changes one thing for every member of the same seasonal forecast; plant density is AquaCrop's only (WOFOST has no plant population). The amounts (100 mm of water, half the soil under residues, ridges that let no rain run off) are assumptions until the site's managers answer the questions below.

## How past forecasts did

Early-maturing maize, FAO agricultural onset: the 1 Oct SEAS5 forecasts of 1993–2016 ranked the years' crops with a correlation of 0.51 (AquaCrop) and 0.54 (WOFOST) against each model's crop in the observed weather; the platform's season-length forecast alone 0.62, its Oct–Dec rain 0.52. 1 is a perfect ranking, 0 none; with 24 years a value is uncertain by about ±0.2.

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First calibration only (maize and sorghum, against trials at Kiboko), not verified against farm yields: for discussion, not advice. The full outlook, its members and the other forecasts: https://operational-multi-model-seasonal-fo.vercel.app/#/crop

Farm managers and agronomists at this site: tell us how much water, residue and ridging farmers use, and the scenarios will be set to match:

https://docs.google.com/forms/d/e/1FAIpQLSeSm0ZhEjV3MhBCayBJEdrwtgLh2iBfyzd-b90o56vXo9B9vw/viewform